SPEECH RECOGNITION

Giving machines the power to understand human speech

1. INTRODUCTION

Speech recognition lets computers identify spoken words and turn them into text or commands. It blends linguistics, acoustics and machine learning to understand how people talk.

"Hello!" → text

2. HOW SPEECH RECOGNITION WORKS

1. Audio Input → **2. Preprocessing** → **3. Feature Extraction** → **4. Acoustic Model** → **5. Language Model** → **6. Decoding** → **7. Output**

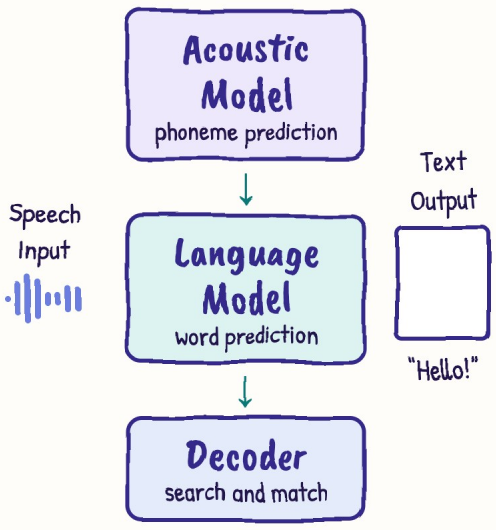
1. **Audio Input** — Speech is captured by a microphone.
2. **Preprocessing** — Noise reduction and framing of the audio.
3. **Feature Extraction** — Key features (e.g., MFCCs) are pulled from the signal.
4. **Acoustic Model** — A neural network predicts the likely sounds (phonemes).
5. **Language Model** — Word patterns and grammar improve accuracy.
6. **Decoding** — Combines both models to find the best word sequence.
7. **Output** — "Hello!" Final text or command is produced.

3. ASR (AUTOMATIC SPEECH RECOGNITION)

ASR is the core technology that automatically converts spoken language into text without human help. Modern systems use deep learning trained on large amounts of transcribed speech.

Key Components

- **Acoustic Model:** maps audio to phonemes.
- **Language Model:** predicts likely word sequences.
- **Decoder:** finds the best match between the two.
- **Training Data:** large sets of speech and transcripts.



4. APPLICATIONS & REAL-WORLD USES

- **Virtual Assistants** — Siri, Alexa and Google Assistant respond to voice commands.
- **Customer Support** — Automatic call transcription and voice helpdesks.
- **Dictation** — Voice typing for documents, emails and notes.
- **Automotive** — In-car voice control for navigation, calls and music.
- **Healthcare** — Doctors dictate patient notes hands-free.
- **Education** — Accessibility tools for students and e-learning.

5. ADVANTAGES AND CHALLENGES

Advantages

- ✓ Hands-free, natural interaction
- ✓ Saves time and boosts productivity
- ✓ Makes devices accessible to more people
- ✓ Works across phones, cars and homes
- ✓ Keeps improving with AI

Challenges

- Accuracy drops with noise and accents
- Needs very large training datasets
- Privacy and security concerns
- Heavy computing and storage needs
- Hard to grasp context and emotion

"Speech recognition bridges the gap between human communication and machine understanding."